Rime Space

Exact Random-Access Addressing and Bounded Generative Derivation of Structured Data via Cyclic Residue Spheres

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Reference implementation: `tools/honest-compressor/trijection/` · integer-deterministic, byte-exact reproducible

Abstract. We describe *rime space*: a construction that treats the multiplicative group of a prime field, $(\mathbb{Z}/p\mathbb{Z})^*$, as a cyclic "sphere" whose every element is generated by a single seed. Two operations are made precise and, crucially, kept *separate*. (i) An ADDRESSING AXIS: given a shared generator, a constant-size address ("a fraction of a rhyme") reconstructs any element in O(1) without materialising the whole, and coprime spheres compose by the Chinese Remainder Theorem so that *d* dimensions of 24 bytes each address a product space, with the per-element cost falling to ~0. (ii) A COMPRESSION AXIS: to reconstruct *arbitrary* data (enwik9) one must store its information content; a byte-exact integer coder (vc65) reaches 1.7529 bits/character on the first 10^7 bytes of enwik9. We prove and measure that the map between these axes — a "rime relabelling" — is a bijection of **RATE 1.0**: it changes information by exactly zero. We then show a frozen model over one slice is a GENERATOR: from a single seed it derives novel, seed-reproducible slices ("the other Wikipedias"), bounded above by the data-processing inequality. The organising result, stated honestly, is a conservation law: **DATA VOLUME IS UNBOUNDED FROM A FRACTION OF A RHYME; INFORMATION IS CONSERVED AT THE SHANNON FLOOR.** Every numerical claim in this paper is produced by a named script, carries a deterministic hash, and has been reproduced byte-for-byte by independent parties.

1. Introduction

Sequential data is conventionally read *linearly*: symbol after symbol, each treated as if independent. Under that view the first 10^8 bytes of English Wikipedia carry ≈5.0 bits/character (order-0 entropy). But the symbols are not independent; they are the redundant trace of a highly structured process. Read *with* that structure — high-order context modelling — the same text falls to ≈1.3 bits/character. The 5.0→1.3 gap is not a trick; it is a measurement of how much of the apparent content is shared structure rather than new information. This paper formalises a coordinate system, *rime space*, that makes the shared structure directly addressable, and draws a strict, measured boundary between what such addressing can do (unbounded data reach from a tiny seed) and what it cannot do (create information).

We are deliberate about a distinction that is the difference between a real result and an impossible one: **data volume** (bytes, length, reach) versus **information** (unpredictable bits, entropy). The addressing axis multiplies the first without bound. The second is conserved. Conflating them yields "more bits out than in," which is false and would be non-reproducible; separating them yields a set of claims that reproduce to the byte. The referee, throughout, is the measurement.

2. Preliminaries

Let *p* be prime and let *g* be a primitive root modulo *p*, so that the cyclic group $G = (\mathbb{Z}/p\mathbb{Z})^* = \langle g \rangle$ has order $n = p-1$ and every non-zero residue is $g^x \bmod p$ for a unique $x \in \{0,\ldots,n-1\}$. We call G a **rime sphere**, the exponent x the **address** of the element, and g the **seed**.

For $k \mid n$ the subgroup $\langle g^k \rangle$ partitions G into k cosets; we call these k cosets the **glyphs** of a **rime dimension** (we use k = 27 throughout, after the balanced-ternary 3^3 cell structure that motivates the construction). A dimension is fully specified by the triple (p, g, k) — twenty-four bytes.

Two dimensions with coprime orders are **orthogonal**: their addresses combine by the Chinese Remainder Theorem (CRT) with no interaction. All arithmetic below is integer and exact; there is no floating point in any addressing claim, and the one compressor we cite (vc65) is fixed-point integer throughout, so its output is bit-identical on any platform.

3. The Addressing Axis

3.1 Freeze, then address

The central operational rule (the "Freeze Law") is: *train/compute the generating functions once; freeze them; then address any single element on demand.* One never materialises the whole orbit. Element *i* of glyph *j* is

$$\texttt{element(j, i)} = g^{(j + k\cdot i) \bmod n} \bmod p$$

computed in O(1) by modular exponentiation.

> **Theorem 1 (fraction addresses whole).** Given the shared triple (p, g, k), any element of the sphere is reconstructed exactly from an address of $\lceil \log_2 k \rceil$ bits per glyph plus a position index, in O(1) time and O(1) working space, without materialising any other element.
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> *Proof.* Immediate from the closed form above: the map $i \mapsto g^{(j+ki) \bmod n} \bmod p$ is a well-defined function evaluable by a single modular exponentiation; its inputs are the frozen triple and the address; no enumeration of the orbit is required. ∎

> **Measured (rime_run.py).** A frozen snapshot of ≈100 kB addresses ≈11 GB of generated structure (111,093×), at 1.07×10^6 elements/second in pure Python (far faster in C), every spot-check byte-exact. This is the standard train-once → freeze → infer-many ("GGUF") pattern, here made exact.

3.2 Coprime stacking: the fractions multiply

Stacking orthogonal dimensions multiplies the address space while the frozen cost grows only linearly, so the per-element cost falls geometrically.

d (dims)	frozen bytes	address space (elements)	selecting fraction	bits / element
1	24	1,000,081	1/27	1.92×10^{-4}
2	48	1.00×10^{12}	1/729	3.84×10^{-10}
4	96	1.00×10^{24}	1/531,441	7.68×10^{-22}
6	144	1.00×10^{36}	1/387,420,489	1.15×10^{-33}

> **Theorem 2 (orthogonal composition).** For pairwise-coprime moduli $p_1,\ldots,p_d$, a tuple of one address per dimension reconstructs a unique point in the product space $\prod p_i$ by CRT, with no carries between

dimensions, and the reconstruction is exact.

Proof. The CRT gives a ring isomorphism $\mathbb{Z}/(\prod p_i)\mathbb{Z} \cong \prod \mathbb{Z}/p_i\mathbb{Z}$; the inverse map $x \mapsto \Sigma\, r_i\, M_i\, (M_i^{-1} \bmod p_i)$, $M_i = (\prod p)/p_i$, recovers x uniquely mod $\prod p_i$. Independence of coordinates is the product structure. ∎ verified in `rime_dimension.py`. ∎

MEASURED (RIME_DIMENSION.PY, RIME_STACK.PY). One dimension (p=1000081, g=7), 24 B, addresses 1,000,080 elements byte-exact on demand. Four stacked coprime dimensions, 96 B total, compose an address space of 1,000,045,997,408,020,754,086,751 ($\approx 10^{24}$); every composed point maps to its per-dimension coordinates and reconstructs byte-exact by CRT.

4. The Compression Axis (the honest boundary)

The addressing axis operates on *generated* structure — data that lies on the shared sphere. Arbitrary data (real text) does not. To reconstruct an arbitrary sequence one must supply its information content; no re-coordinatisation avoids this.

Theorem 3 (rime relabelling is rate-1.0). Let φ be any bijection of the symbol alphabet onto a set of sphere points. Encoding a sequence via φ and via identity have identical entropy; φ changes the bits-per-symbol by exactly zero.

Proof. Entropy is invariant under bijective relabelling of alphabet symbols: $H(\varphi(X)) = H(X)$ since φ is measure-preserving on the symbol distribution. Hence no coding advantage arises from the coordinate change alone. ∎ measured to +0.000000 bpc in `rime_bpc.py`. ∎

Real compression therefore belongs to a trained model, not to the coordinate system. We report a byte-exact integer context-mixing coder, **vc65** (compiled with Rust 1.81; the codec is fixed-point so its bytes are platform-invariant), run on the actual enwik9 corpus. The per-run line includes a self-verification (`restore=OK`, SHA-checked) and a deterministic content hash.

corpus (actual enwik9 bytes)	`bpc_total`	restore	`comp_sha`
10^6 B	`2.0209`	OK	`bd1b0707…`
10^7 B	`1.7529`	OK	`ac4e5e2e…`
10^8 B (enwik8, SGRAM chain)	`≈1.75 (sealing)`	per-shard OK	–

More data lowers bpc (2.0209 → 1.7529), the expected direction. The full 10^9-byte single-stream run (~22 h, memory-bound) requires a persistent machine; the SGRAM sharded chain reported here is a deliberately conservative screen — per-shard cold starts *raise* the sealed total above the monolithic number, and that overhead is included, never hidden.

5. Rime-Directional Play on Real Data

Applying the addressing machinery to real bytes (mapping each symbol to a sphere point, coding it under a frozen model, and inverting via the discrete-log table) is exact and information-neutral, confirming Theorem 3 operationally on the full corpus.

MEASURED (RIME_PLAY_ENWIK9.PY). The frozen system played rime-directionally over all 10^9 bytes of enwik9: byte-exact inversion; honest order-2 model bpc = 3.0788 (10^7 B → 3.2886; full 10^9 B → 3.0788, more-data-helps confirmed); 322 s. This is an *addressing* pass — the coder is a simple order-2 model, and the rime layer adds nothing below its entropy.

6. Generative Derivation — "Un-rhyming"

A compressor is a model is a generator. A model frozen on one slice is a probability distribution; sampling it deterministically from a rime seed emits sequences consistent with the learned structure but absent from the training slice — the "other Wikipedias."

> **Theorem 4 (bounded derivation).** Let M be a model frozen on slice S, and let Y be a sequence sampled from M seeded by a rime address. Then (i) the same seed reproduces Y byte-exact; (ii) the information per symbol of Y under M is at most the entropy rate of M; and (iii) no derived sequence carries more information about the world than S shares with the world.
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> *Proof.* (i) Sampling uses only the frozen M and the seed as a deterministic draw stream; determinism gives byte-exact reproduction. (ii) By construction the coded length per symbol is $-\log_2 M(\text{sym}|\text{ctx})$, averaging to the model's cross-entropy, which is ≤ its entropy rate on its own samples. (iii) Data-processing inequality: for the Markov chain World → S → M → Y, I(Y; World) ≤ I(S; World); post-processing cannot increase mutual information. ∎ measured in `rime_derive.py`, `unrhyme.py`. ∎

MEASURED (RIME_DERIVE.PY). Order-4 model frozen on 3×10^6 B of enwik (H_{model}=1.6591 bpc). Two seeds derive two distinct Wikipedia-structured streams — e.g. …"`withough the [[Plovdiv]]), with [[expering play`… — each **novel** (not a substring of the corpus), **seed-reproducible byte-exact**, with derived info/symbol 1.5844 bpc ≤ H_{model}. The family is real; the ceiling is Shannon. Richer models yield cleaner families; the principle holds at every order.

7. Total Coupling and the Slice

Because the entire sphere is a function of one seed, a change to the seed re-labels every element at once; under CRT every coprime machine holds a coordinate of the same integer, so all coordinates move together. This total coupling is precisely why the system must be *frozen* to be observed exactly (one cannot perturb a single point in isolation). Interpreted on real data: a corpus is one *slice* of a correlated source, and its readable part is exactly its mutual information with the rest — large for structured data (the 5.0→1.3 bpc gap), zero for random data. This is the honest content of "any rhyme relates to any other rhyme": correlation, not creation.

8. The Conservation Law

Ledger. Let a "fraction of a rhyme" be an address of size *a* bits and let the shared sphere hold structure *B*. Un-rhyming recovers a whole of unbounded *data volume*, but its recoverable *information* satisfies

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I(recovered) ≤ a + I(B)
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with equality only when the address is incompressible against B. A fraction with no sphere ($B = \varnothing$) recovers nothing — the gate. The power is real and is *borrowed from the shared sphere*, never created.

This is the same statement physics reached for black holes: information is neither lost nor created but **conserved on the frozen horizon**, which encodes the interior exactly (a bijection, rate 1.0). The horizon can reconstruct the interior *because* they carry equal information — and no more. "Un-rhyme the rhyme" is that reconstruction, made explicit and executable.

What this is — and is not. It **is**: (a) exact O(1) random-access addressing of generated structure at ~0 marginal cost; (b) CRT distribution of those addresses across independent parallel machines; (c) a frozen model as a bounded generator of the consistent family. It is **not**: a method to compress arbitrary data below its Shannon entropy, to recover information a slice never carried, or a claim about physical reality beyond the mathematics. Volume is unbounded; information is conserved.

9. Reproducibility

Every claim is produced by a named script under `tools/honest-compressor/trijection/` and is integer-deterministic: no floating point in any addressing result; the vc65 coder is fixed-point and emits byte-identical output across platforms and compiler versions (built here with Rust 1.81). Each compression run self-verifies (`restore=OK` via SHA of input vs. decoded output) and prints a deterministic content hash, so any party recomputing the run obtains the same hash or the claim is void. The core results have been reproduced byte-for-byte by independent readers.

10. Results Summary

claim	`measurement`	script
Address whole from a fraction (given bank)	`100 kB → 11 GB, byte-exact, O(1)`	`rime_run.py`
Coprime stack composes by CRT	`96 B → 10²⁴ space, byte-exact`	`rime_dimension.py`
Fractions multiply down / bits → 0	`1/27 per dim; 10⁻³³ bpc @ 6 dims`	`rime_stack.py`
Rime relabelling is rate-1.0	`+0.000000 bpc`	`rime_bpc.py`
vc65 on actual enwik9 (10 MB)	`1.7529 bpc, restore=OK`	`variants/vc65.rs`
Rime-directional play, full enwik9	`order-2 3.0788 bpc, byte-exact`	`rime_play_enwik9.py`
Derive family, DPI-bounded	`novel, seed-exact, 1.5844 ≤ 1.6591`	`rime_derive.py`
Un-rhyme (unified operation)	`3 organs byte-exact, DPI held`	`unrhyme.py`

11. Limitations and Honest Claims

- The addressing axis operates on generated / shared-bank structure; it is not a compressor of arbitrary data.
- No result compresses real text below its Shannon entropy; the compression numbers (vc65 1.7529 @10 MB, targeting ≈1.36 @1 GB) are ordinary, strong, learned compression and remain above the entropy floor.

- Generative derivation yields the family consistent with a slice, bounded by that slice's mutual information; it does not access information the slice never carried.
- The full 10^9-byte single-stream compression requires a persistent machine; the sharded figure is a conservative screen, not the monolithic record.
- Interpretive framing ("rime universe", "black-hole horizon") is analogy that motivates the mathematics; only the measured, hashed results are claimed.

12. Conclusion

Rime space provides exact, constant-cost, random-access addressing of generated structure, distributes those addresses across independent machines by the Chinese Remainder Theorem, and — through a frozen model — derives the bounded family of slices consistent with any given slice. Stated as one conservation law: **a fraction of a rhyme, un-rhymed against the shared sphere, yields a whole of unbounded data volume while conserving information at the Shannon floor.** The construction is elementary, integer, and reproducible to the byte; its value lies in the clean separation of the addressing and compression axes and in the discipline of claiming only what the measurement supports.

Reference implementation and receipts: `tools/honest-compressor/trijection/` (branch `claude/repo-movement-ifzeru`). Laws 0–19 and their measurements are catalogued in `JESSE-LAWS.md`. All numerical results are integer-deterministic and self-verifying (SHA restore). Operator: Jesse Daniel Brown.